Financial Econometrics

Panel Data Analysis

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Does managerial ownership affect firm value?

As we already discussed in the Lab Session of Week 6 - Non-linear functional form, managerial ownership might affects firm value. Please see the corresponding Lab session, for the economic rational. In this lab session, we investigate if we find stronger or weaker support of a positive effect if exploiting a panel set-up.

Please use *MB_Own.Rdata*, a data-frame containing following variables:

- *YEAR* is fiscal year
- *GVKEY* is company identifier
- *mtbe* is market-to-book
- *own* is CEO ownership

Task 1: Prepare the data

- Load the data
- Remove missing values using *na.omit()*
- Remove erroneous entries (e.g., negative *mtbe*)
- Winsorize *own* and *mtb* at the 99% level
- Declare it to be a panel data. Please see the coding hint and see *?pdata.frame*:

```
# package
require(plm)

# declare data-frame as pdata-frame
... <- pdata.frame(..., index = c("...", "..."))
```

Task 2: Equivalence of "first difference", "binary regressors", and "entity-demeaned" approach using only the years 2000 and 2008

- Limit the data to the years $t = 2000$ and $t = 2008$ and firms i with complete data in both years
- Estimate following benchmark model

$$mtbe_{t,i} = \alpha + \beta_1 \times own_{t,i} + u_{t,i}$$

- Estimate following model that has firm fixed effects α_i

$$mtbe_{t,i} = \alpha_i + \beta_1 \times own_{t,i} + u_{t,i}$$

  using three different estimation techniques:

  - First difference approach: *fit = plm(y~0 + x , data = ..., model = "fd")*. Please note the “0” indicating that the constant will be dropped. Otherwise the coefficients would not be identical.
  - Binary regressor: *fit = plm(y~x+ factor(firm identifier), data = ..., model = "pooling")*
  - Entity-demeaned approach: *fit = plm(y~x , data = ..., model = "within", effect = "individual")*.
- Please cluster standard errors by firm for all regressions, except for model 3, “binary regressors”. While you should always cluster standard errors, you might run out of memory for model 3, hence drop it for now.

```r
require(car)
require(lmtest)
require(sandwich)

# clustered standard errors
coeftest(fit, vcov = vcovHC(fit, cluster = "group", type = "sss")
```

- Make a stargazer table summarising the output. Use *stargazer(..., omit = "GVKEY")* to avoid that the table contains all the fixed effects from the binary regressor model.
- Does it matter if we include firm fixed effects or not? Why?

Task 3: Full sample estimation

Please estimate following models using the full sample:

- Estimate following benchmark model

$$mtbe_{t,i} = \alpha + \beta_1 \times own_{t,i} + u_{t,i}$$

- Estimate following model that has time fixed effects λ_t

$$mtbe_{t,i} = \lambda_t + \beta_1 \times own_{t,i} + u_{t,i}$$

  - using a time-demeaned approach: *fit = plm(y~x, data = ..., model = "within", effect = "time")*.

- Estimate following model that has firm fixed effects α_i and time fixed effects λ_t

$$mtbe_{t,i} = \alpha_i + \lambda_t + \beta_1 \times own_{t,i} + u_{t,i}$$

  using two different estimation techniques

  - Entity-demeaned approach with explicit year fixed effects: *fit = plm(y~x + factor(YEAR), data = ..., model = "within", effect = "individual")*.
  - Entity-demeaned and time-demeaned approach: *fit = plm(y~x , data = ..., model = "within", effect = "twoway")*.

- Make a stargazer table summarising the output. Use *stargazer(..., omit = "YEAR")* to avoid that the table contains all the fixed effects from the Binary Regressor model.

- Does it matter if we include firm fixed effects or not? Why?